

Dart Tutorial 6: Strings in Dart

This chapter explains how to work with text data in Dart using Strings. After completing this tutorial, students will understand how to create, combine, and manipulate Strings using built in methods.

1. Introduction to Strings

A **String** in Dart is a sequence of characters used to represent text data.

Strings are one of the most commonly used data types since almost every app deals with text such as names, messages, and labels.

2. Why We Use Strings in Dart

In real world apps, we constantly deal with text data such as usernames, messages, and titles.

Strings allow us to:

- Store and display text based information.
 - Combine multiple pieces of text together.
 - Search, modify, and format text easily.
 - Validate user input such as emails or passwords.
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3. Creating Strings

Strings in Dart can be created using single quotes or double quotes.

```
String name = 'Ishfaq';  
String course = "Flutter Development";  
  
print(name);  
print(course);
```

Output:

```
Ishfaq  
Flutter Development
```

4. String Concatenation

Concatenation means joining two or more Strings together.

```
String firstName = 'Code';  
String lastName = 'With Ishfaq';  
  
String fullName = firstName + ' ' + lastName;  
print(fullName);
```

Output:

```
Code With Ishfaq
```

5. String Interpolation

Instead of using `+` to combine variables, Dart allows us to insert variables directly inside a String using `$`.

```
String name = 'Ishfaq';  
int age = 22;  
  
print('My name is $name and I am $age years old');
```

Output:

```
My name is Ishfaq and I am 22 years old
```

For expressions, curly braces are used:

```
print('Next year I will be ${age + 1}');
```

Output:

```
Next year I will be 23
```

6. Multi-line Strings

Dart allows writing Strings across multiple lines using triple quotes.

```
String message = '''
This is line one
This is line two
''';

print(message);
```

7. Common String Methods

Method	Description	Example
toUpperCase()	Converts to uppercase	'dart'.toUpperCase() → DART
toLowerCase()	Converts to lowercase	'DART'.toLowerCase() → dart
trim()	Removes extra spaces	' dart '.trim() → dart
length	Returns string length	'dart'.length → 4
contains()	Checks if text exists	'dart'.contains('ar') → true
replaceAll()	Replaces text	'dart'.replaceAll('d', 'c') → cart

```
String text = ' Learn Dart ';

print(text.trim());
print(text.toUpperCase());
print(text.length);
```

8. Splitting Strings

The `split()` method breaks a String into a list based on a separator.

```
String sentence = 'Dart is fun to learn';
List<String> words = sentence.split(' ');

print(words);
```

Output:

```
[Dart, is, fun, to, learn]
```

9. Checking String Properties

```
String text = '';

print(text.isEmpty);
print(text.isNotEmpty);
```

Output:

```
true
false
```

10. Conclusion of Tutorial 6

In this tutorial, students have learned:

1. What Strings are and why they are important.
2. How to create Strings using single and double quotes.
3. How to combine Strings using concatenation and interpolation.
4. How to use multi-line Strings.
5. How to use common String methods like `toUpperCase`, `trim`, and `split`.
6. How to check if a String is empty or not.

In the next tutorial, we will learn about Lists in Dart.

This completes Dart Tutorial 6.